Suggested FYP topics for 2026-27
by
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Project code :	CN1
Project title :	Campus Light Bus Navigation Web Application
No. of students :	1-2
Description of project :	CUHK campus operates a paid light bus service for visitors, but first-time visitors often lack clear information on which stop to get off for specific destinations and may miss their stops. This project aims to design and implement a web application that provides real-time bus location, helps users select their intended destination from a drop-down menu, identifies the correct stop, and issues alerts as the bus approaches it. The system should also provide walking directions from the bus stop to the final destination within campus, integrating with campus map data where appropriate. The student will be responsible for the full development cycle: requirements gathering, system design, implementation of the web front-end and back-end, and testing with actual campus users. Prior experience with web development is helpful but not mandatory; the student should be willing to learn relevant technologies during the project.
Industrial collaboration :	No.

Project code :	CN2
Project title :	Information Theoretic Inequalities in Network and Related Areas
No. of students :	1-2
Description of project :	This is a research-oriented project on proving information theoretic inequalities in collaboration with the supervisor and/or doctoral students. Possible directions include problems at the intersection of network information theory (e.g., interference channels, multi-user networks, capacity/outer bounds), additive combinatorics (e.g., sumset inequalities, entropy analogues, formal equivalence results), and functional analysis (e.g., Gaussian extremality, variational methods, De Bruijn and Fisher information inequalities). The precise topic will be tailored to the student's background and interests, with an emphasis on developing a deep understanding of the underlying theory and techniques. The student will study selected research papers, work through detailed proofs, and aim to obtain new or sharpened inequalities or structural insights in one of the above areas. This project is suitable for students with strong mathematical maturity who are interested in pursuing research or further studies in information theory or related fields.
Industrial collaboration :	No.